

ECE 315 Lab Report Guidelines

General Information

As part of the ECE315 course, students are required to submit four formal lab reports. These are to be completed collaboratively with your assigned lab group (unless prior approval is given to work individually). These reports are essential to fulfilling the lab component of the course.

Please adhere to the format outlined below for each report. Additionally, specific lab assignments may include unique questions or requirements detailed in the lab handout's report requirements section.

Report Format

Each report must include:

- **Cover Page:** Must include the course number, lab section number, lab number, due date, and names of all contributing students. For privacy reasons, student ID numbers are not required.
- **Table of Contents:** Optional, but recommended for reports exceeding ten pages.
- **Abstract:** A concise, one-page summary including:
 - Paraphrased lab objectives.
 - Summary of utilized hardware.
 - Overview of completed exercises.
 - Design Section: A detailed description of your solutions to the lab exercises, including alternative solutions explored. Merely repeating the specifications is insufficient.
- **Test Suite:** You must explain how you tested and validated your system. Describe your testing approach, how you verified correctness, and how you handled edge or boundary cases. The emphasis is on demonstrating a clear and systematic testing process, not just reporting that tests passed.

Present your test suite in table format, including:

- Test description
- Input conditions (if applicable)
- Expected outcome
- Actual result
- Pass/Fail status

- **Results:** Incorporate all required results as specified in the lab handout, such as tables and comparisons.
- **Discussion:** Interpret your results in this section, discussing how they relate to your hypothesis and to the background context provided in the introduction. Consider:
 - Analyzing discrepancies: If there were unexpected results, discuss possible reasons.
 - Relating to previous research: Compare your findings with those of similar studies.
 - Implications: What do your results imply for the field of study?
- **Conclusion:** Evaluate whether the lab's objectives were achieved and summarize your findings. Discuss any factors contributing to incomplete or unsuccessful aspects of the lab.

Coding Style

All submitted C code must follow good programming practices and maintain a consistent style. Code quality is part of your evaluation.

General Requirements:

- Use meaningful and descriptive variable and function names. Avoid single-letter variable names except for loop counters.
- Properly align indented blocks, braces and maintain consistent formatting throughout the file.
- Clear and concise comments.
- Initialize variables before use.
- Group related declarations together.
- Use constants (`#define` or `const`) instead of hardcoded “magic numbers.”
- Handle edge cases and invalid inputs where applicable.
- Avoid duplicated code; use functions to promote reuse.
- Poorly formatted, difficult-to-read, or poorly structured code may result in point deductions even if the program functions correctly.

Grammar and Spelling

Reports must be checked for spelling and grammar. Reports with significant errors will incur penalties.

Report Appearance

Reports must be typed; handwritten submissions will not be accepted.

Use a reader-friendly font size and style; avoid fonts smaller than size ten.

Diagrams can be hand-drawn but should be neat and utilize a straight edge for clarity.

Format source code for readability, ensuring comments do not disrupt the flow of text.

Report Length

There is no strict length requirement; however, reports should be concise, including only necessary information. Both excessively lengthy and incomplete reports will face deductions, whereas succinct and comprehensive reports will be rewarded.

Report Submission

You must submit **one single zip file** containing:

1. **The final lab report in PDF format**
2. **All source files modified by your group**
 - Include only the files you modified.
 - Do NOT submit unaltered files.
 - Do NOT submit the entire workspace or full lab folder.

Failure to follow submission instructions may result in penalties.

These guidelines are designed to help you produce high-quality reports that reflect your understanding and engagement with the lab material. Adherence to these guidelines will contribute to your success in the lab component of ECE 315.